



1.2 Exponents and logarithms

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Exponent laws: $a^m \cdot a^n = a^{(m+n)}$, $a^m / a^n = a^{(m-n)}$, $(a^m)^n = a^{(mn)}$, $a^0 = 1$, $a^{(-n)} = 1/a^n$. Logarithms are the inverse of exponentiation: $\log_a(x) = y$ means $a^y = x$. Key laws: $\log_a(xy) = \log_a x + \log_a y$, $\log_a(x/y) = \log_a x - \log_a y$, $\log_a(x^n) = n \cdot \log_a x$, and the change of base rule $\log_a x = \ln x / \ln a$.

Worked example: follow the reasoning

Solve $2^{(3x-1)} = 40$ for x .
Take $\ln$ of both sides: $(3x-1)\ln 2 = \ln 40$.
 $3x-1 = \ln 40 / \ln 2 = 5.3219$
 $3x = 6.3219$
 $x = 2.107$ (3 d.p.)

Quick check · multiple choice

Simplify $\log_3(9) + \log_3(3)$.

- A 2
- B 3
- C 4
- D 9

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Solve $\log_2(x) + \log_2(x-2) = 3$ for x , stating any restriction on x . [7]
(b) Solve $5^{(2x)} = 3^{(x+1)}$ giving x to 3 significant figures.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $\log_2[x(x-2)]=3$ so $x(x-2)=8$, $x^2-2x-8=0$, $(x-4)(x+2)=0$, $x=4$ or $x=-2$. Since $x>2$ required (domain), $x=4$.

(b) $2x \ln 5 = (x+1) \ln 3 \rightarrow 2x \ln 5 - x \ln 3 = \ln 3 \rightarrow x(2 \ln 5 - \ln 3) = \ln 3 \rightarrow x = \ln 3 / (2 \ln 5 - \ln 3) = 1.0986 / 2.1188 = 0.519$.

Quick check answer

B. $\log_3(9)=2$ and $\log_3(3)=1$, so the sum is 3 (equivalently $\log_3(27)=3$).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.